

JAGIRYALA DEEPAK GOUD

 github.com/deepakgoud0  linkedin.com/in/deepakgoudjagiryala0212  deepakgoud1979@gmail.com  +91-9390148199

 portfolio-lu2q.vercel.app

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 2+ years of experience building and scaling multi-tenant SaaS applications. Skilled in **JavaScript, TypeScript, Node.js, React, MongoDB, and AWS**, with hands-on experience in **third-party API integrations, cloud deployment, authentication and security, caching and database performance optimization, and real-time systems**. Strong foundation in **data structures, algorithms, object-oriented programming, and system design**.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, GLSL

Frontend: React (React 19), Next.js, Vite, Tauri, TanStack Query/Router, Jotai, Radix UI, Tailwind CSS, Three.js (react-three-fiber), Recharts, HTML5, CSS3

Backend: Node.js, Bun, Elysia, REST APIs, WebSockets, Webhooks, SOAP

Databases: MongoDB, Mongoose, MongoDB Compass, Redis, PostgreSQL (basic)

Cloud & DevOps: AWS (EC2, S3, SES, SNS, Cognito, Secrets Manager, CloudWatch), Docker, Vercel, Render, PM2, Linux, CI/CD, GitHub Actions

Auth & Security: Auth0, AWS Cognito, JWT, OAuth

Observability & Testing: Sentry, Pino (structured logging), Bun Test

Concepts: Data Structures, Algorithms, OOP, System Design, REST API Design, Multi-tenant SaaS, Caching Strategies

Tools: Git, GitHub, Postman, MongoDB Compass, VS Code

PROFESSIONAL EXPERIENCE

IVECTORS SOFTWARE SOLUTIONS LLP

Software Engineer (Full Stack)

Jun 2024 – Present

Hyderabad, India

- Built and maintained scalable backend services for a multi-tenant SaaS healthcare platform using Node.js, TypeScript, Bun, and MongoDB.
- Integrated 6+ third-party services across REST, SOAP, webhook, and SFTP protocols, normalizing heterogeneous data formats (JSON, XML).
- Designed and built a custom parser and encoder for a complex structured data-interchange format (EDI X12), handling encoding, decoding, validation, and envelope generation.
- Optimized MongoDB aggregation pipelines and indexing (profiled with MongoDB Compass) and added Redis caching to reduce dashboard and query load times.
- Implemented authentication and authorization using Auth0, AWS Cognito, and JWT/OAuth across a multi-tenant environment.
- Deployed and operated services on AWS (EC2, S3, SES, SNS, Cognito, CloudWatch) with PM2 process management, and built scheduled cron jobs for automated reporting and data synchronization.

- Implemented real-time data synchronization using WebSockets and webhooks, and set up Sentry error monitoring with structured logging (Pino) for production reliability.
- Developed responsive frontend features in React (React 19) using TanStack Query/Router, Jotai, Radix UI, and Tailwind CSS, including data-visualization dashboards; managed Git/GitHub branching, code reviews, and release workflows.

PROJECTS

Snip — URL Shortener with Analytics

2026

Bun, Elysia, TypeScript, MongoDB, Redis, React [\[Live\]](#) [\[Code\]](#)

- Built a full-stack URL shortener with a Bun/Elysia TypeScript API, MongoDB persistence, and **Redis-cached redirects** (1-hour TTL) with hit/miss counters, serving lookups from cache instead of the database.
- Implemented **HS256 JWT authentication from scratch** (HMAC signing with constant-time signature comparison) and argon2 password hashing, with owner-scoped route guards for per-user link access.
- Built a click-analytics dashboard using **MongoDB aggregation pipelines** — clicks over time, top referrers, device breakdown, and live cache-hit ratio — with React 19, TanStack Query, and Recharts.
- Deployed end-to-end: Dockerized API on Render, frontend on Vercel, MongoDB Atlas and Upstash Redis, with environment-based CORS locking.

Interactive 3D Universe Portfolio

2026

React, TypeScript, Three.js, GLSL, WebGL [\[Live\]](#) [\[Code\]](#)

- Built an explorable 3D universe in React and Three.js (react-three-fiber), rendering **109,400 real stars** from the HYG catalogue packed into a 1.7 MB binary (16 bytes/star) and drawn as GPU point sprites in a single draw call.
- Implemented a **real-time gravitational-lensing black hole in GLSL** by numerically integrating null geodesics per pixel, producing an accretion disk with relativistic Doppler beaming and a photon ring at 60 FPS.
- Rendered volumetric nebulae via ray-marched density fields, and modelled the Milky Way, Local Group, and cosmic web at true astronomical scale using discrete level-of-detail scenes.
- **Code-split each scene into lazily-loaded chunks, cutting the initial bundle from 707 KB to 394 KB gzipped**; optimized textures and deployed on Vercel.

Live Weather Application

2024

JavaScript, REST API, Bootstrap

- Real-time weather app using the OpenWeatherMap API with asynchronous data fetching and a responsive UI.

EDUCATION

RGUKT Basar

2020 – 2024

B.Tech – Metallurgical and Materials Engineering

CGPA: 6.35